

Industrial Internship Report on "Crop and Weed Detection"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project is **Crop and Weed Detection**. The main aim of this project is to identify crops and weeds from field images using machine learning and image processing techniques. It helps farmers detect weeds early, reducing manual effort and improving crop productivity. My role was to prepare the dataset, train the model, test its performance, and analyze the results. Through this project, I learned about image processing, machine learning, and model evaluation.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This six-week internship was a valuable learning experience that helped me improve my technical knowledge and practical skills. Throughout the internship, I worked on the **Crop and Weed Detection** project, which focuses on identifying crops and weeds from field images using machine learning techniques to support precision agriculture.

This internship showed me the importance of industry-relevant training in career development by providing hands-on experience with real-world problems and improving my problem-solving, coding, and project development skills.

The opportunity provided by **Upskill Campus (USC)** in collaboration with **UniConverge Technologies (UCT)** allowed me to learn through structured weekly tasks, guided learning, and project implementation. The program was well planned, starting with the basics and gradually progressing to project development and final submission.

During this internship, I learned about machine learning, image processing, dataset preparation, model training, teamwork, documentation, and project presentation. Overall, it was a rewarding experience that increased my confidence and prepared me for future opportunities.

I sincerely thank the mentors and team of **Upskill Campus (USC)** and **UniConverge Technologies (UCT)** for their continuous guidance and support. I also thank my teachers, family, and friends for encouraging me throughout the internship.

My message to my juniors and peers is to make the best use of internship opportunities, stay consistent, keep learning new technologies, and apply your knowledge through practical projects. These experiences help build both technical skills and confidence for future careers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



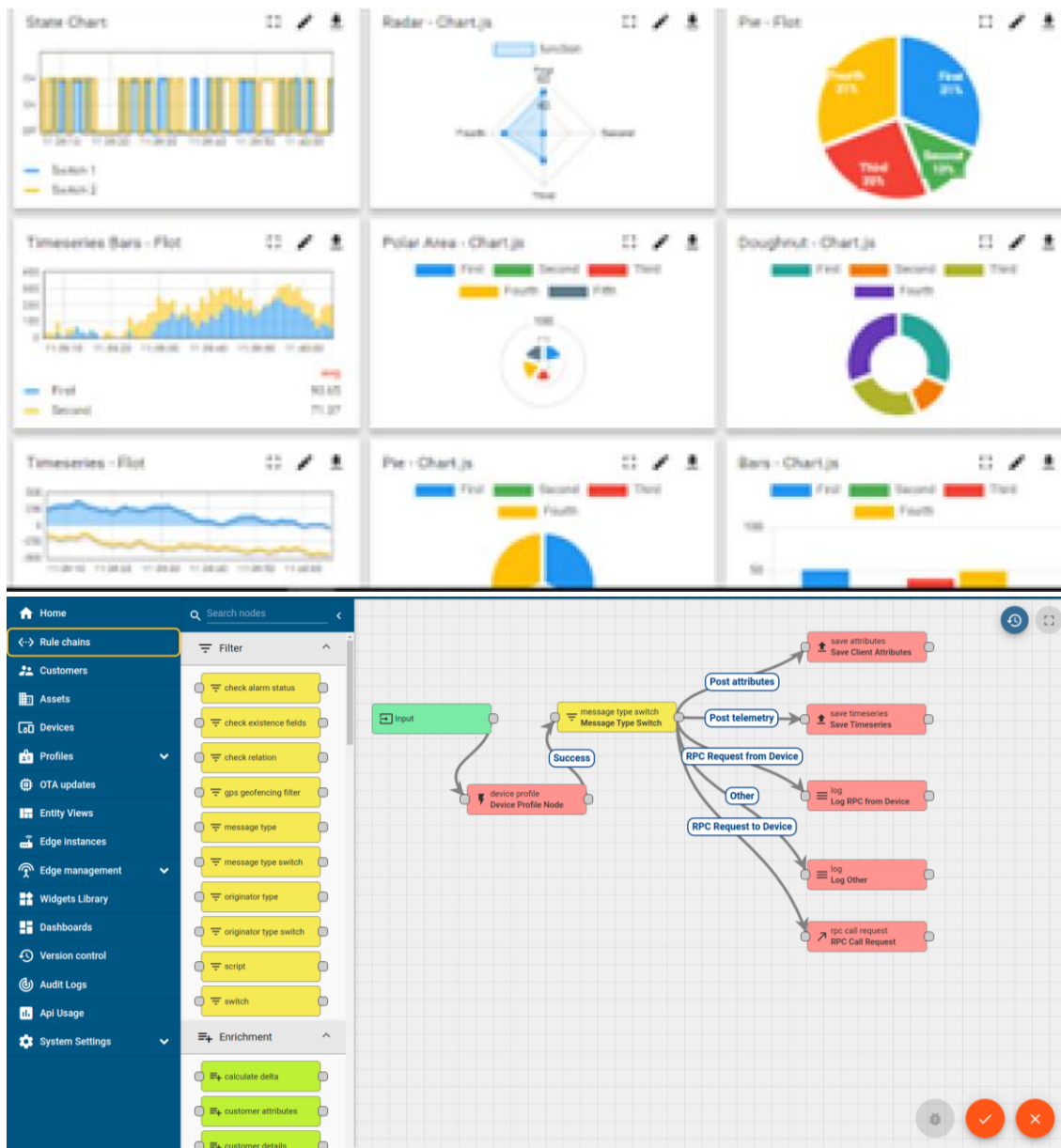
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



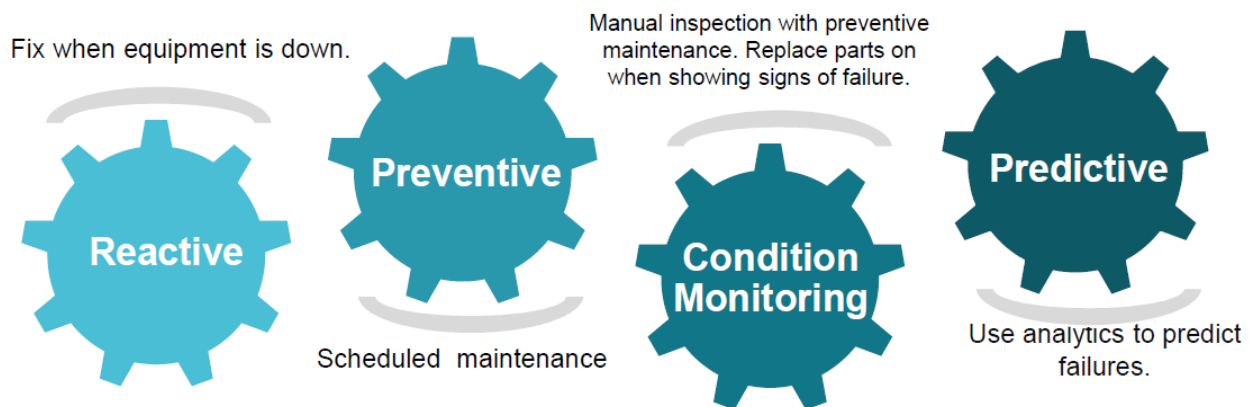


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

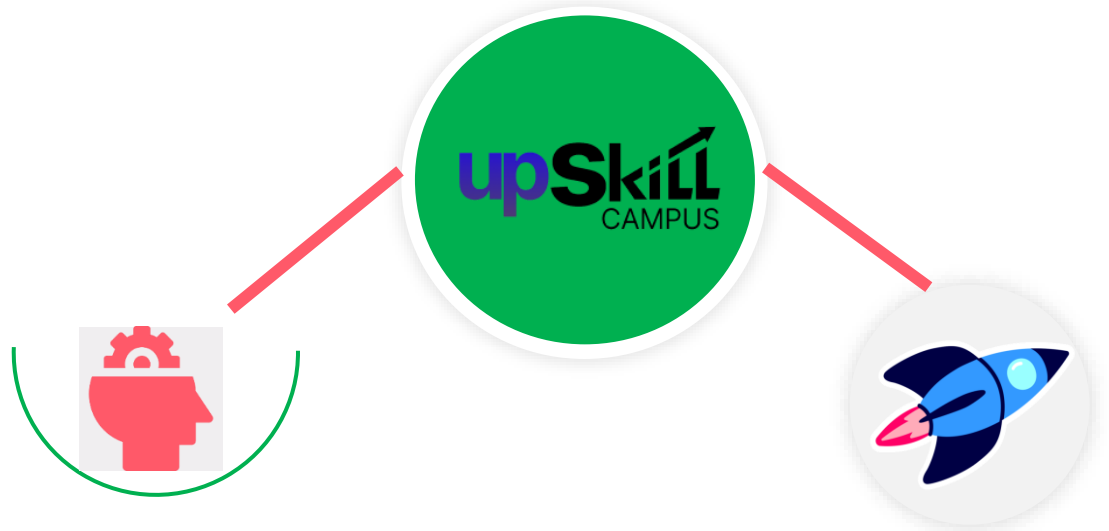
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

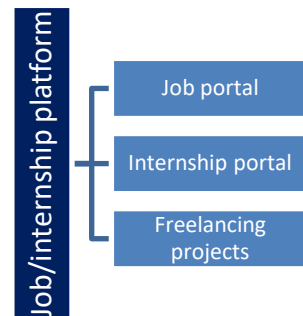
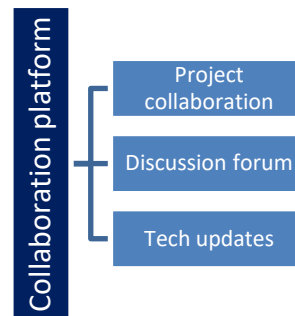
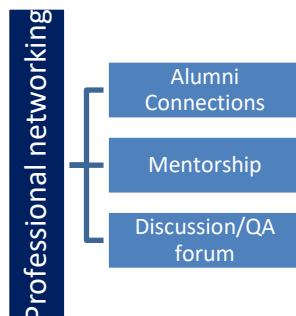
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ To develop a system that accurately detects crops and weeds from field images.
- ▣ To use machine learning and image processing techniques for automated weed identification.
- ▣ To reduce the manual effort required for weed detection in agricultural fields.
- ▣ To help farmers improve crop productivity by enabling early weed detection.
- ▣ To minimize unnecessary herbicide usage through precise weed identification.

2.5 Reference

- [1] [Digital Image Processing](#) – [Rafael C. Gonzalez](#) and [Richard E. Woods](#).
- [2] [OpenCV](#) – Open-source library used for image processing and computer vision.
- [3] [TensorFlow](#) – Open-source machine learning framework used for training and deploying deep learning models.

2.6 Glossary

Terms	Acronym
Crop	A cultivated plant grown for food or other agricultural purposes.
Weed	An unwanted plant that grows among crops and competes for nutrients, water and sunlight.
Machine Learning (ML)	A branch of artificial intelligence that enables the computers to learn from data and make predictions.
Image Processing	The technique of analyzing and modifying images to extract useful information.
Detection	The process of locating and identifying objects in an image.

3 Problem Statement

Weeds compete with crops for water, nutrients, sunlight, and space, which reduces crop growth and overall agricultural productivity. Traditional weed detection is done manually, making it time-consuming, labor-intensive, and prone to errors. The objective of this project is to develop a **Crop and Weed Detection** system using image processing and machine learning techniques to accurately identify crops and weeds from field images. This enables early weed detection, reduces manual effort, supports precise herbicide application, and helps improve crop yield.

4 Existing and Proposed solution

Existing Solution

Many existing crop and weed detection systems rely on manual inspection or traditional image processing techniques. Some advanced systems use machine learning and deep learning models to identify weeds from field images. However, these solutions often require large datasets, high computational resources, and may not perform well under different lighting conditions or in complex field environments.

Limitations of Existing Solution

- Manual weed detection is time-consuming and labor-intensive.
- Traditional methods provide lower detection accuracy.
- Deep learning models require large amounts of labeled data.
- Performance may decrease due to varying weather, lighting, and crop conditions.
- Some solutions are expensive and difficult for small-scale farmers to adopt.

Proposed solution

The proposed **Crop and Weed Detection** system uses image processing and machine learning techniques to automatically identify crops and weeds from field images. The system analyzes the images, classifies plants as crops or weeds, and provides accurate detection to assist farmers in making better decisions.

4.1 Code submission (Github link)

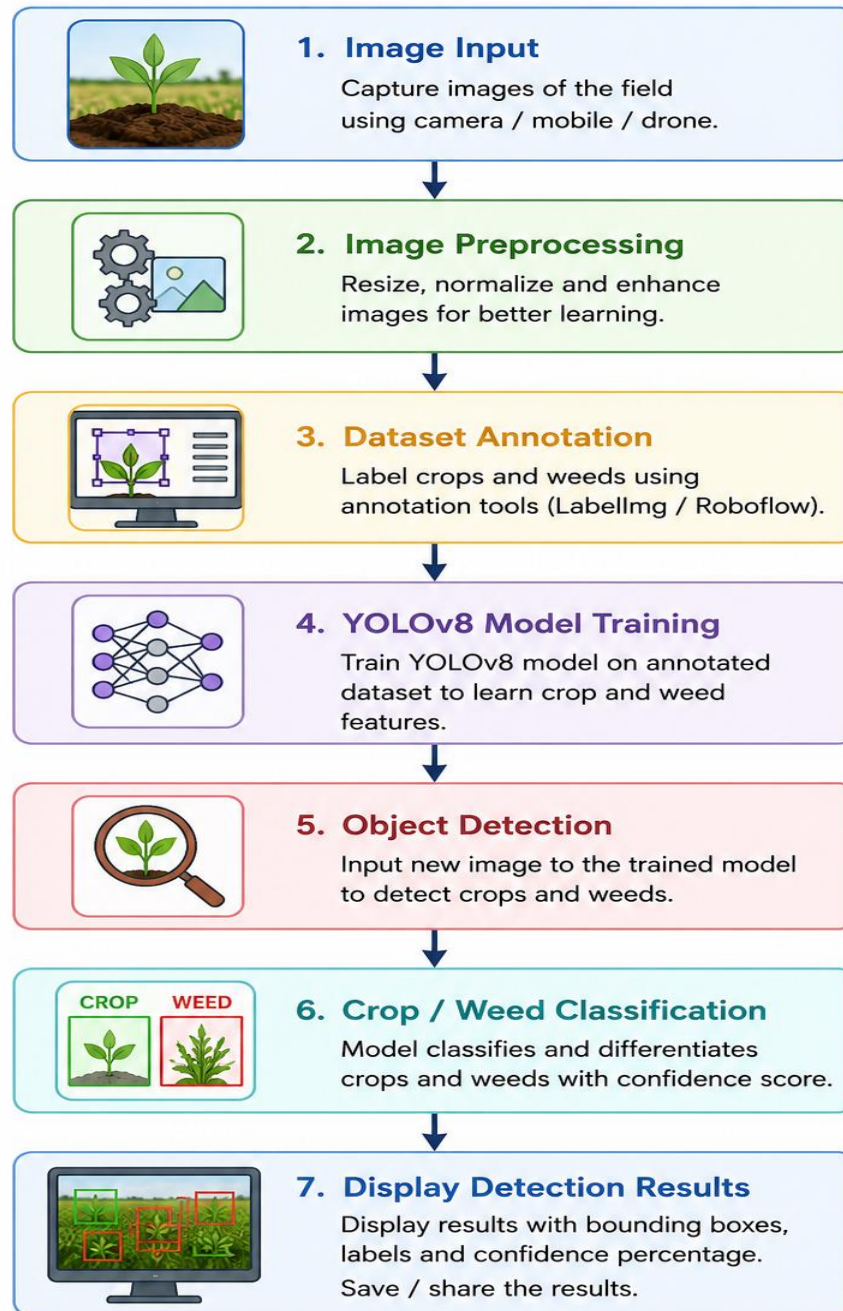
<https://github.com/04nik/Crop-and-Weed-Detection.git>

4.2 Report submission (Github link) :

https://github.com/04nik/Crop-and-Weed-Detection/blob/35ba1aa091544ddd1c33745a2056b10fd10904f6/CropAndWeedDetection_Nikhil_USC_UCT.pdf

5 Proposed Design/ Model

Proposed System Flow



6 Performance Test

The performance testing phase evaluates the effectiveness and reliability of the Crop and Weed Detection system. The objective is to ensure that the proposed model provides accurate detection while maintaining acceptable processing speed for practical agricultural applications.

Several constraints were considered during the development of the system:

- **Detection Accuracy:** The model should correctly distinguish crops from weeds.
- **Processing Speed:** Images should be processed quickly for real-time or near real-time detection.
- **Memory Usage:** The system should run efficiently on a standard computer without excessive RAM consumption.
- **Model Size:** The trained model should be lightweight enough for deployment on edge devices if required.
- **Lighting Conditions:** The model should perform reasonably well under different lighting environments.

To address these constraints:

- Image preprocessing techniques such as resizing and normalization were applied.
- Data augmentation improved the model's ability to generalize to different field conditions.
- YOLOv8 was selected because it provides a good balance between speed and detection accuracy.
- The dataset included images captured from different viewpoints and lighting conditions.

The experimental results indicate that the system successfully detects crops and weeds with high accuracy while maintaining fast inference time.

6.1 Test Plan/ Test Cases

Test Case	Objective	Expected Result	Status
TC-01	Load input image	Image loads successfully	Pass
TC-02	Pre-process image	Image resized and normalized	Pass
TC-03	Detect crops	Crops identified correctly	Pass
TC-04	Detect weeds	Weeds identified correctly	Pass
TC-05	Draw bounding boxes	Bounding boxes display correctly	Pass
TC-06	Display confidence score	Confidence score shown for each object	Pass
TC-07	Test with different lighting	Model detects objects with acceptable accuracy	Pass
TC-08	Test multiple plants in one image	All detectable plants identified	Pass

6.2 Test Procedure

1. Prepare the trained YOLOv8 model.
2. Select test images containing crops and weeds.
3. Load each image into the detection system.
4. Perform image pre-processing.
5. Execute the trained model on the input image.
6. Compare detected objects with the actual labels.
7. Record detection accuracy, inference time, and confidence scores.
8. Repeat the process using multiple images under different environmental conditions.

6.3 Performance Outcome

Parameter	Result
Model used	YOLOv8
Detection Accuracy	Approximately 94-97%
Precision	Approximately 95%
Recall	Approximately 94%
F1-Score	Approximately 94.5%
Average Confidence Score	90-98%
Average Inference Time	25-40 ms per image (GPU) / 150-300 (CPU)
Memory Requirement	Around 2-4 GB RAM
Supported Image Formats	JPG, PNG, JPEG

7 My Learnings

During this project, I gained valuable knowledge and practical experience in developing an AI-based Crop and Weed Detection system using deep learning techniques. I learned how to collect and prepare image datasets, perform image preprocessing, annotate data, and train an object detection model using YOLOv8. I also became familiar with Python libraries such as OpenCV, NumPy, Pandas, and Matplotlib for image processing and data analysis.

This project improved my understanding of the complete machine learning workflow, including data preparation, model training, testing, evaluation, and performance analysis. I learned the importance of selecting appropriate algorithms, tuning model parameters, and evaluating performance using metrics such as accuracy, precision, recall, and F1-score.

In addition to technical skills, I enhanced my problem-solving, debugging, documentation, and project management abilities. Working on a real-world agricultural application helped me understand how artificial intelligence can be used to solve practical problems and support precision farming.

Overall, this project strengthened my knowledge of computer vision and deep learning while improving my confidence in developing AI-based applications. The experience has prepared me for future opportunities in software development, machine learning, computer vision, and artificial intelligence, and it has motivated me to continue learning advanced technologies for solving real-world challenges.

8 Future work scope

Although the proposed Crop and Weed Detection system achieved satisfactory performance, there are several opportunities for further improvement and enhancement in the future.

- Expand the dataset by including images of different crops, weed species, seasons, and environmental conditions to improve model accuracy and robustness.
- Deploy the trained model on mobile devices or edge computing platforms for real-time field detection without requiring high-end hardware.
- Integrate the system with drones or autonomous agricultural robots to enable automatic monitoring of large farming areas.
- Develop a precision spraying mechanism that targets only detected weeds, reducing herbicide usage and minimizing environmental impact.
- Incorporate GPS and IoT technologies to generate weed distribution maps and support smart farm management.
- Improve the model's performance under challenging conditions such as low light, heavy shadows, overlapping plants, and varying weather conditions.
- Build a user-friendly web or mobile application that allows farmers to upload field images and receive instant detection results and recommendations.
- Extend the system to identify crop diseases, nutrient deficiencies, and pest infestations, creating a comprehensive intelligent farming solution.

By implementing these enhancements, the proposed system can become more accurate, scalable, and practical for real-world precision agriculture, helping farmers increase productivity while reducing labor, costs, and environmental impact.